SMART TRAVEL PLANNER

Project Report

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Fundamentals in AIML

1. Abstract

The Smart Travel Planner is a computer program that helps people plan trips. It uses Artificial Intelligence to figure out where people might want to go based on how they're feeling and how much money they have to spend. The Smart Travel Planner also thinks about why someone's traveling.

It is like having a person to help plan trips. It is a computer program. The Smart Travel Planner gives people a lot of ideas for places to go which makes it really helpful. The Smart Travel Planner is an useful tool for planning trips because it can think like a person and give people many options, for travel destinations.

2.Introduction

When we think about traveling there are many tourism options to consider. Choosing the place to go has become really tough. This project wants to make things easier for us by using Artificial Intelligence concepts.

The system asks for our travel ideas. Then suggests places that are just right for us kind of like a real person who helps us plan our trip, a human travel advisor. The system is like a travel helper that uses our inputs to find the travel destinations for us just like a human travel advisor would do, for us.

3. Objectives

- To design a smart travel recommendation system
- To apply Artificial Intelligence concepts practically
- To develop a user-friendly graphical interface
- To provide personalized travel suggestions

4. LITERATURE REVIEW

When we think about travel systems we usually have to look for places to go and then decide where we want to go.. Now that we have something called Artificial Intelligence it is easier for computers to suggest places for us to visit.

Artificial Intelligence systems that use rules to make decisions are really simple. These systems work by following rules that people have already set up. The project I am working on uses this rule-based approach to suggest travel destinations to people who are looking for places to visit. The travel destinations are recommended using a rule-based system that's a type of Artificial Intelligence.

5. METHODOLOGY

Step 1: Input Collection

The system collects user inputs:

- Mood
- Budget
- Purpose

Step 2: Knowledge Base

A structured database (places_db) stores travel destinations categorized by:

- Mood
- Budget
- Purpose

Step 3: Processing

- The system applies rule-based logic
- Matches user input with database entries
- Selects suitable destinations

Step 4: Output Generation

- Displays 2–3 recommended places
- Results shown in GUI

6. SYSTEM DESIGN

Input → Process → Output Model

Input: User preferences
↓
Processing: Rule-based AI system
↓
Output: Travel destination suggestions

7. TECHNOLOGIES USED

- Programming Language: Python
- Library: Tkinter (GUI)
- Concepts: Artificial Intelligence (Rule-Based System)

8. IMPLEMENTATION

The system is implemented using Python.

- A dictionary is used as a knowledge base
- Tkinter is used to design GUI
- Random module is used to provide multiple suggestions

9. FEATURES

- User-friendly GUI
- Dark theme interface
- Multiple mood categories
- Large database of destinations
- Smart recommendation system

10. INPUT PARAMETERS

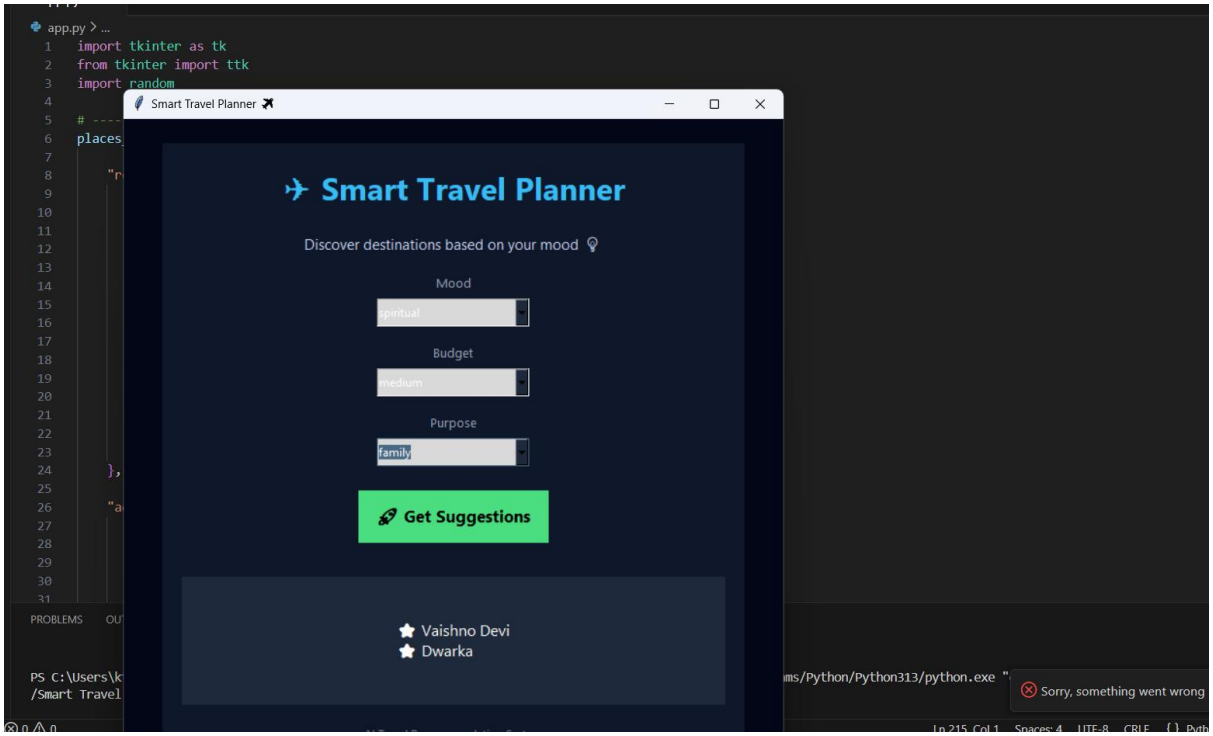
- Mood: Relax, Adventure, Romantic, Spiritual, Party, Nature
- Budget: Low, Medium, High
- Purpose: Solo, Family, Friends

11. OUTPUT

The system provides a list of recommended travel destinations based on user input.

Example:

- Rishikesh
- Kasol
- Dharamshala



12. RESULT

The system successfully generates relevant travel suggestions based on user preferences. The GUI enhances usability and makes the system interactive.

13. FUTURE SCOPE

- Integration with maps (A* algorithm)
- Adding real-time cost estimation
- Image-based recommendations
- Mobile application development

14. CONCLUSION

The Smart Travel Planner demonstrates how Artificial Intelligence can be used in everyday applications. The system effectively recommends travel destinations using a rule-based approach and provides a user-friendly interface.

16. REFERENCES

- Artificial Intelligence by Stuart Russell & Peter Norvig
- Python Documentation
- Tkinter Documentation

17. AUTHOR DETAILS

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